LAB 6:
DYNAMIC ROUTINGCONFIGURATION:RIP, EIGRP, OSPF, BGP

OBJECTIVES

a) To study and understand the operating concepts of dynamic routing protocols such as RIP, EIGRP, OSPF, and BGP.
b) To implement and configure these routing protocols on routers using Cisco Packet Tracer in order to allow automatic route exchange and successful communication between multiple networks.

HARDWARE AND SOFTWARE REQUIREMENTS

1. Cisco Packet Tracer
2. Windows PC/Laptop

THEORY

1. RIP (Routing Information Protocol)

RIP is a distance-vector routing protocol that determines the best path based on hop count. It operates by periodically sending the complete routing table to directly connected neighboring routers every 30 seconds. The maximum allowable hop count in RIP is 15, and any route exceeding this value is considered unreachable. Due to its simple design and slow convergence, RIP is mainly suitable for small networks.

2. EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP is an advanced routing protocol developed by Cisco and is often referred to as a hybrid routing protocol. It uses the Diffusing Update Algorithm (DUAL) to select the most efficient path to a destination. Unlike RIP, EIGRP sends only partial updates when network changes occur, which helps reduce bandwidth usage. It is commonly used in Cisco-based enterprise networks that require fast convergence and reliable performance.

3. OSPF (Open Shortest Path First)

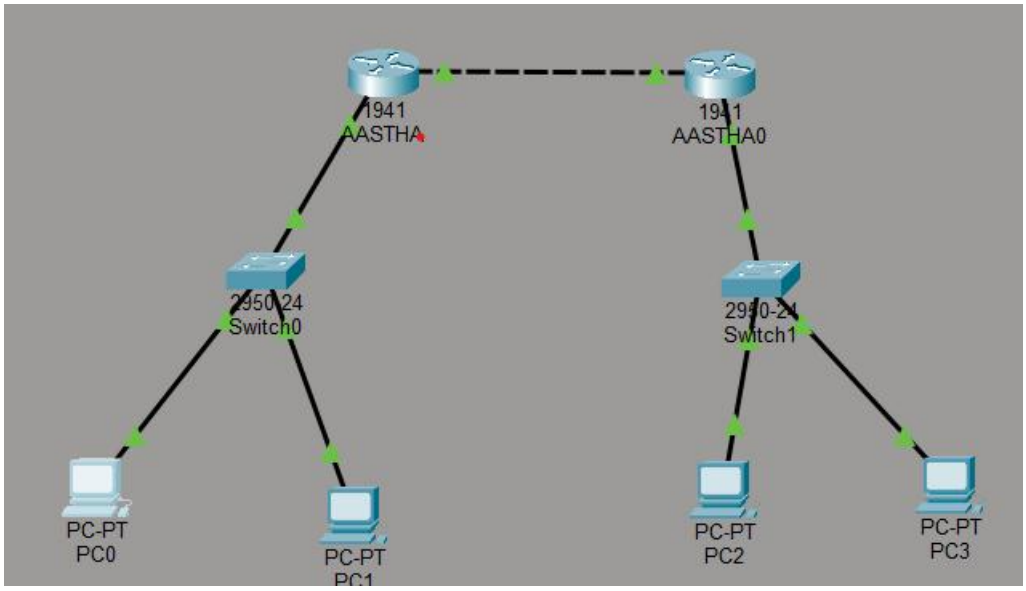
OSPF is a link-state routing protocol that maintains detailed information about the entire network topology. It uses the Dijkstra Shortest Path First (SPF) algorithm to calculate the optimal route to each destination. Routers exchange Link State Advertisements (LSAs) to build and maintain a topology database. OSPF is widely used in large and scalable networks because it supports hierarchical network design using multiple areas, with Area 0 as the backbone.

4. BGP (Border Gateway Protocol)

BGP is a path-vector routing protocol designed to exchange routing information between different autonomous systems. Instead of relying only on metrics such as distance or cost, BGP makes routing decisions based on network policies and administrative rules. It is the main protocol used for routing traffic between Internet Service Providers and plays a critical role in the functioning of the global Internet

NETWORK TOPOLOGY

A topology was created using Router0(AASTHA) and Router1(AASTHA0) connected via a serial/WAN link. Switch0 connects PC0 and PC1 to Router0 (Network 192.168.1.0/24). Switch1 connects PC2 and PC3 to Router1 (Network 192.168.2.0/24).



Configuration Table:

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Gig0/1	192.168.1.1	255.255.255.0	N/A
Router0	Gig0/0	10.0.0.1	255.0.0.0	N/A
Router1	Gig0/1	192.168.2.1	255.255.255.0	N/A
Router1	Gig0/0	10.0.0.2	255.0.0.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1

PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

1. RIP Configuration Commands

Router0:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Router1:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

2. EIGRP Configuration Commands

Router0:

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

Router1:

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.2.0 0.0.0.255
```

```
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

3. OSPF Configuration Commands

Router0:

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

Router1:

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

4. BGP Configuration Commands

Router0 (AS 65001):

```
Router(config)# router bgp 65001
Router(config-router)# neighbor 10.0.0.2 remote-as 65002
Router(config-router)# network 192.168.1.0 mask 255.255.255.0
```

Router1 (AS 65002):

```
Router(config)# router bgp 65002
Router(config-router)# neighbor 10.0.0.1 remote-as 65001
Router(config-router)# network 192.168.2.0 mask 255.255.255.0
```

RESULT

The network was successfully configured using the various dynamic routing protocols. To verify the connectivity, a ping request was sent from PC0 (192.168.1.2) connected to Router0 to PC2 (192.168.2.2) connected to Router1. For each protocol configured (RIP, EIGRP, OSPF, and BGP), the routing tables were updated automatically, and all packets were sent and received successfully which validates the connection is correct.

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126
Reply from 192.168.2.2: bytes=32 time=12ms TTL=126
Reply from 192.168.2.2: bytes=32 time=4ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 6ms
```

Fig: Ping in RIP configuration

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126
Reply from 192.168.2.2: bytes=32 time=12ms TTL=126
Reply from 192.168.2.2: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 8ms
```

Fig: Ping in EIGRP

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

Fig: Ping in OSPF

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=12ms TTL=126
Reply from 192.168.2.2: bytes=32 time=10ms TTL=126
Reply from 192.168.2.2: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 12ms, Average = 11ms

C:\>
```

Fig: Ping in BGP

DISCUSSION AND CONCLUSION

During the lab session, we explored and implemented various dynamic routing protocols to understand how routers automatically share routing information. Through RIP, we learned the fundamentals of distance-vector routing, while EIGRP helped us observe faster convergence and efficient path selection. OSPF provided insight into link-state routing and the use of topology databases, whereas BGP introduced the concept of routing between different autonomous systems.

Thus, the lab was successfully completed with a clear understanding and practical implementation of dynamic routing protocols using Cisco Packet Tracer.